From Feedback to Revision: When Do Multi-Agent Roles Look Load-Bearing?

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Abstract

Multi-agent LLM systems are specified as role diagrams — Planner, Executor, Reviewer — and the diagram is read as a mechanism. We ask when the roles are load-bearing. Holding agents, topology, call budget, evaluator and grading fixed, we delete every role instruction and, separately, enforce the same division of labor at the interface, across three actor models (11,600 frozen trajectories), plus an exploratory post-freeze role-free condition that receives the evaluator's feedback.

Deleting the roles destroys the intended division of labor on Omni-MATH — the rate at which every agent asserts its own final answer rises from 0.000–0.120 to 0.846–0.976 — while first-pass accuracy does not fall. Roles look load-bearing for repair only against a role-free team whose prompts never carried the evaluator's feedback (gemma-4: 0.222 with roles, 0.051 without). Given the same feedback string, the role-free team recovers more: 0.609 against 0.222 for gemma-4 (89 jointly failed problems recovered only without roles, 4 only with them; exact McNemar $p = 6.2 \times 10^{-22}$), and reaches higher final accuracy in all five actor–domain cells. Yet when both teams are credited with the same number of verdicts per problem, the role-free advantage disappears: the pooled difference is +0.009 in favor of roles (95% CI −0.014 to +0.033); no robustness check favors the role-free team significantly, and one favors roles. Roles mainly change how often a team asks for feedback. The role-structured Reviewer withholds approval: on Omni-MATH, gemma-4's team submits 1.27 times per failed trajectory against 3.20. Overriding the Reviewer in an exploratory 400-problem run, role prompts unchanged, raises gemma-4's recovery from 0.198 (gated re-run) to 0.613, matching the role-free team's 0.549 on the same problems. **Structure is not adaptability**: a clean division of labor is not evidence that a team acts on corrective information, and role ablations must hold information channels fixed, not only budgets.

Keywords

Multi-agent systems, large language models, role specialization, division of labor, organizational structure, ablation

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1 Introduction

A multi-agent LLM system is usually presented as a diagram: a Planner that decomposes without solving, an Executor that carries out the plan, a Reviewer that critiques rather than re-derives (PER), and arrows for who hands what to whom [18, 33, 39]. The diagram is read as a *mechanism*: it says where the system's competence comes from, and why one pays for three agents instead of one. What runs, however, is three prompts handed to a language model, which may honor the division of labor, ignore it, or satisfy its surface while doing the whole task in every box. An accuracy score cannot tell these apart: a pipeline whose three agents have quietly become three copies of the same solver returns the same number as one whose organization works exactly as drawn.

The question. We ask what a diagram cannot answer: **for which phase of the work, and under what conditions, is multi-agent organization load-bearing?** We ask per phase because a team solving a problem does two different things: it produces an answer and, if that answer comes back wrong, it must produce a *different* one. A component is **load-bearing** if removing or altering it reproducibly degrades task performance *under matched call and information budgets* (§3). Matching information is the subtle part: feedback reaches each agent through its prompt, and the prompt is exactly what a role ablation rewrites.

Removing roles reorganizes the team; the first answer is no worse. Deleting every role instruction — with topology, actor model, retry budget and evaluator held fixed — destroys the division of labor in all three actors: on Omni-MATH the rate at which every agent asserts its own final answer goes from 0.000–0.120 to 0.846–0.976 (the role-free prompt asks each agent for its current answer). First-pass accuracy does not fall (unchanged for gemma-4, +0.05 for gpt-oss), in line with prior role ablations [10, 31]. Read alone, this is the familiar negative result: the organization was decorative.

Roles look load-bearing for repair, until the feedback is matched. After a rejected first answer, roles do seem to matter: the role-structured team recovers 0.222 of gemma-4's failures, a role-free team 0.051. But the role-structured templates carry the evaluator's verdict and repair hint to every agent, and the role-free template has no slot for them. Given the same feedback string — an exploratory condition added after the snapshot was frozen — the role-free team recovers *more*: on the 236 problems both teams first answer wrongly it recovers 89 that the role-structured team misses, against 4 the other way (exact McNemar $p = 6.2 \times 10^{-22}$), and its final accuracy is higher in every actor and on 574 free-response college-science problems. **Roles appear load-bearing for repair only when their prompts are the only channel carrying the feedback.**

Roles decide when the team asks. The role-free team with feedback also asks for more of it. Credited with only as many evaluator

verdicts as the role-structured team received on the same problem, it is no better: recovery differs by +0.009 pooled, in favor of roles (95% CI −0.014 to +0.033), and a role-structured re-run made after the freeze shows no significant difference either. Per verdict, roles add little; what they mainly change is how often a team asks for feedback. The role-structured Reviewer withholds approval before any feedback exists and, on Omni-MATH, mostly where no clause forbids it: gemma-4's team submits 1.27 times per failed trajectory against 3.20, and 84 of its 316 failed trajectories never submit within the six-cycle budget. Overriding the Reviewer in a post-freeze run on 400 problems, with the role prompts unchanged, raises gemma-4's recovery from 0.198 (gated re-run) to 0.613, as high as the role-free team's (0.549): against a ground-truth evaluator and a capped budget, a withheld submission is a lost chance to recover. Enforcing the division of labor at the interface loosens the gate and raises recovery in every actor (resolvably only for Llama-3.3), but not end-to-end accuracy: 8–17% of enforced trajectories never finish.

Contributions. (1) An operational definition of load-bearing, applied per phase through a budget-matched design that varies organization in both directions. (2) Evidence that a role ablation which drops the feedback channel can manufacture an apparent role effect on repair (clearly for gemma-4, directionally for gpt-oss), and that, across three actors on competition mathematics and two on college science, the role-free advantage disappears once the feedback string and the number of verdicts are matched. (3) An account of what roles do instead: a gate on submission, measured from the logs and tested by removing it. (4) A measurement point that extends a recent separation of reviewer precision from critique uptake [40]: properties that look like good collaboration — a precise reviewer, a clean division of labor — are not evidence that corrective information changes the next action (§6).

2 Related Work

The question predates the LLM literature that reopened it. Organization-centered multi-agent systems gave roles a formal existence independent of their occupants: Moise$^+$ separates a structural, a functional and a *deontic* specification — what a role-filler is obliged and permitted to do [21] — OperA makes role adoption an explicit social contract [7], and electronic institutions bind norms through middleware outside the agents [11], which can regiment a norm (make violation impossible) or enforce it (sanction violation) [17]. An LLM role prompt is the degenerate case: functional and deontic clauses written in prose, with no middleware to make them bind and no record of whether they did. In these terms our role-free condition keeps the structure — order, graph, a positional approver — and deletes the functional and deontic specifications together; our enforced condition regiments the functional part at the interface. "Adaptability" here means revising an answer within an episode, not organizational reorganization [19]; teamwork models treat repair of a failed joint plan as an obligation of the team [37], which a prompt can state but not secure.

Role specialization and role drift. That prompted roles are not reliably obeyed is established: Cemri et al. [4] name disobeying the role specification a failure mode across 1,600+ annotated traces, Cao et al. [3] give the training-time account of *role drift*, Orogat et al. [31] find role-label conditioning does not induce the intended specialization, Patel [32] measures collapse as near-identical rationales, Kandoussi [25] reports differentiation arising *without* assigned roles, and enforcement-side work makes role scope structural rather than advisory [13, 36]. Role collapse, and the decoupling of role quality from task performance, are known [3, 40]; we measure them under a controlled ablation.

Redundancy and parity. Dylan et al. [10] hold total LLM calls fixed and find an evolved Planner–Executor–Critic team no better than an evolved single agent ($p = 0.80$). We replicate first-pass parity only in part: gpt-oss *improves* by +0.0503 with an interval excluding zero. Zhang et al. [43] argue that gains of multi-agent efficiency methods often arise where one agent suffices — an objection aimed at our role-free condition, which is why we report a single-agent reference (§4); Chen et al. [6] prune redundant agents on the same premise.

Feedback, revision and answer lock-in. Iterative refinement with self-generated or tool feedback is a standard pattern [15, 28, 34], as is multi-agent debate [9, 27]. That models repeat earlier outputs under exact verifier feedback is established [22]; under self-reflection the failure has a name, Degeneration-of-Thought [27], and belief entrenchment is studied directly [29]. Huang et al. [20] are frequently misread here: their *oracle* condition, which uses the correct label as a stopping rule, *improves* accuracy (75.9 → 84.3 on GSM8K) and only the *intrinsic* condition degrades [see also 24, 35]. Our loop is oracle-gated like theirs, and with a verifier in the loop more attempts buy more coverage [1] — the regime in which our role-structured team's restraint costs it. Closest to our lever is Chen et al. [5], who hold an erroneous claim byte-identical and vary only the *role label* it carries, moving explicit-correction rate by 23–93 percentage points — strong evidence that a role label is causal for error correction, but single-agent and without budget matching. We add the multi-agent case: a role-free team largely repeats itself when the verdict does not reach it and revises when it does. The repetition follows the information, not the organization.

Coordination versus capability. Canedo [2] frames structured specification as a *capability equalizer* whose value falls with model strength; [42] reaches the same prediction from theory and Dochkina [8] the complementary case. Our enforced condition reproduces that direction *only under a scoring rule that drops* 8–17% *of trajectories*; under our pre-registered rule it reverses for gemma-4 and vanishes for the others (§5.5).

Prior analyses of a released corpus, and critique uptake. Three studies analyze a publicly released multi-agent trace corpus on this benchmark [23, 40, 41], from which our single-agent comparator is also drawn. The closest separates reviewer *precision* from critique *uptake* [40]: a planner–executor–reviewer pipeline's reviewer flags real errors more precisely than broadcast-style peer discussion does, yet its correct critique far less often changes the next candidate, and uptake improves when critique is placed in the solver's working context. We ask the organizational counterpart under controlled conditions: with the corrective signal held fixed, does the role structure itself make a team more adaptive? **None of these studies manipulates the role structure**, measures whether the

specified division of labor is realized, or matches feedback access across conditions.

What is open. These literatures establish that roles are often not obeyed, that deleting them leaves first-pass accuracy intact, that models fail to revise under external feedback, and that structure tends to help weaker models. Each concerns a *different stage of the work*, and the failure to revise is almost always measured in a single agent, where there is no organization to ablate. No study holds them in one frame.

3 Definitions and Measurement

3.1 What "load-bearing" means here

An organization is specified by a tuple: agents, an interaction order, a message-passing graph, a context-visibility policy, and a *role contract* per agent. A contract asserts that an agent performs some function (MUST decompose the problem) or that it does not (MUST NOT state a final answer); clauses of the second kind make the specification a *division* of labor. Because a clause is prose handed to a language model, the *realized* organization is not guaranteed to match it, and nothing records which clauses were honored.

Definition. A component of an organization is **load-bearing** if removing or altering it reproducibly degrades task performance *under matched call and information budgets*. The budget clause is what makes it usable: removing roles together with agents, calls or context tests scale, not organization. Because performance itself decomposes — producing a first answer, repairing a wrong one — the definition is applied *per phase*. The information clause covers every channel through which a signal reaches an agent; in a prompted system feedback arrives through the same templates that carry the roles, so a role ablation can change both, and a contrast is matched only where it changes one (§4).

3.2 Measuring the realized organization

Functional collapse (primary). A first pipeline pass *collapses* if *all three* agents independently assert their own final answer, making the pipeline three solvers rather than a division of labor. It targets the property the diagram asserts rather than what a hash can see, is measured on the first pass so it is not confounded with trajectory length, and is not applicable to the structurally enforced conditions, whose schema forbids an answer field.

Exact output duplication (secondary, a lower bound). Two or more role messages byte-identical, as `sha256` of the raw completion. Precision 1.000, recall 0.262 against the model adjudication (§5.1): it shows that at least a given amount of collapse occurred and is never used to compare actors.

Answer authorship, and computation. Whether a Planner authors a final answer — the clause structural enforcement makes unrepresentable, and the manipulation check. Separately, whether a Planner abstains from *computing*, scored as containing no evaluated numeric equality (**P1a**, pre-registered). Being the absence of a pattern, P1a rewards short messages, so it cannot measure a division of labor and we use it only as a fidelity check (§4.1).

First pass and recovery. First-pass accuracy grades the first answer submitted to the evaluator, which need not come from the first pipeline pass: for the role-structured team it may follow internal revision, never external feedback; a trajectory that never submits is graded once, on its final answer, when the budget runs out, and that grade is its first pass. *Recovery* is the fraction of trajectories whose first graded answer was wrong (*failed* trajectories) that nonetheless finish correct. It conditions on a post-treatment variable, so recovery contrasts are descriptive. The feedback-matched contrast is also reported on unconditional final accuracy (§5.2); the verdict-matched analysis (§5.3) is conditional by construction.

Validation. P1a was hand-adjudicated against a definition frozen *before* the adjudication sheet was opened and clears a pre-set agreement threshold of 0.80 (supplement).

4 Intervention

We vary organization in both directions from one baseline: we remove it, and we enforce it. All conditions share three agents, the same interaction order (Planner → Executor → Reviewer), message-passing graph, context visibility (except under enforcement, below), total call budget, verifier and verifier prompt, retry budget, decoding parameters (temperature 0, identical `max_tokens`), stopping rule (a PASS ends the trajectory) and grading. Every trajectory in the frozen snapshot carries the identical budget record: the design tests organization, not scale. **Besides the roles, three things differ.** After a FAIL the evaluator returns a repair hint, written with the gold answer in view and instructed not to reveal it; the role-structured and enforced templates show it to all three agents in the next cycle, each through its role's advice slot. The role-free template has no such slot, so the role-free condition is reported both without and with the feedback. The role-free footer asks every agent for an answer, and the role-structured Reviewer carries two clauses the role-free prompt lacks (below).

Role-structured PER (baseline). Role prompts as ordinarily written: the organization as specified, including a Reviewer told to approve only a complete answer of the right shape and, in a cycle that receives a FAIL, to route the work back to the Planner or Executor rather than approve.

Role-free, feedback withheld. Identical topology and budget; all three agents get one byte-identical prompt: "You are one of three agents collaboratively solving this problem. Continue the work," the problem, the shared history, the agent's slot number, and a footer asking for its current best answer in `\boxed{}` and a closing `[Agree]` (ready to submit) or `[Continue]` — the conventions the role-structured Executor and Reviewer use. Removed is every clause saying who does what, above all the Planner's MUST NOTs on computing and answering. In every prose condition the third agent's `[Agree]` is approval (untagged counts as `[Continue]`) and the latest boxed answer is submitted. Nothing in this template carries the evaluator's verdict or hint.

Role-free with feedback. The same role-free prompt, byte for byte, plus the evaluator's verdict and repair hint — the string the role-structured team receives — shown to all three agents as an `[Evaluator]` entry after the shared history in the cycle after a

FAIL, the lifetime the advice slot has. It differs from the feedback-withheld one only in feedback, and from the role-structured one in roles, in the Reviewer's clauses and in feedback framing.

Structurally enforced. Specialization is enforced by the interface rather than by wording. The Planner emits JSON against a schema with *no answer field*, so a final answer is unrepresentable; the Executor receives the parsed plan object and the problem — never the Planner's raw text — and emits `{derivation, answer}`; the Reviewer emits `{verdict, target_span, critique}`, and `accept` promotes the Executor's answer verbatim *by reference*, having no field in which to *author* one. Each function stays with one agent under the same budget, but the Executor sees less context and a rejected emission can exhaust the regeneration budget. A *context-restored control* repeats it with shared history restored.

Single-agent reference. A budget-matched single-agent protocol on the same problems, from a previously released corpus rather than this campaign (§4.2), guards against "the system was one agent all along" [43]. It cuts both ways: gemma-4's role-structured and feedback-withheld role-free teams finish below their single agent (−0.0225, −0.0563) and gpt-oss's feedback-withheld role-free team above it (+0.0294); the arms come from separate runs (supplement).

4.1 Instrumentation and pre-registered analyses

An intervention that never engaged cannot explain an outcome, so engagement is measured rather than assumed: every rejected-and-regenerated emission is logged with its reason, the enforcement rate counts role-boundary violations only, and answer-bearing calls are counted per condition. One instrumentation gap escaped this discipline (§5.1).

All five endpoints were fixed in writing and committed before any intervention data existed; the full table, with what happened to each, is in the supplement. Units are (problem, actor) pairs and all conditions run the same problems, so every test is paired with bootstrap clustering by problem. **The pre-registered primary endpoint is null**: no pairwise difference-in-differences of P4 — whether enforcement's effect on accuracy differs by actor — survives Holm (smallest adjusted $p = 0.17$; before correction the Llama-3.3−gemma-4 interval, +0.040 [+0.001, +0.080], only just excludes zero, and a separate resample gives $p = 0.055$), and only the gpt-oss−gemma-4 contrast was powered for an interaction of plausible size. The P1a manipulation objective — enforcement driving Planner computation-abstention to ≈1.0 — also failed: the schema forbids an *answer*, not a computation. It was re-designated to answer authorship. **The role-structured-versus-role-free contrast this paper is organized around was pre-registered as a secondary, exploratory replication of a published null**; splitting it into phases is post hoc, and the role-free condition with feedback was added after the snapshot was frozen, with its design and primary analysis committed in writing before any of its data existed. Its results are exploratory. Exclusions were fixed in advance, including the rule that decides §5.5: *unparseable trajectories count as incorrect rather than dropped.*

4.2 Campaign

Task and problems. Omni-MATH competition mathematics [12], sampled by a deterministic seeded manifest so every prefix of the run is a valid random sample; all conditions run the same problems, so every comparison is paired. A multiple-choice benchmark was ruled out before launch: on a four-option item "produce a different candidate" succeeds one time in three by chance.

Actors and scale. Three actor models served from one pinned vLLM endpoint [26]: gpt-oss-120b [30], gemma-4-31B [14] and Llama-3.3-70B [16]. The primary analysis reads a single frozen, sha256-manifested snapshot: 11,600 trajectories over 83,320 role messages and 17,486 gradings, on 1,600 problems sampled uniformly at random from the 4,181 available. The three main conditions run $n = 1{,}600$ per cell for gemma-4 and gpt-oss and $n = 400$ for Llama-3.3; the context-restored control covers gemma-4 and gpt-oss at $n = 400$ each. The evaluator is gpt-oss-120b in every cell, itself one of the actors under study (§7). The single-agent comparator is *not* part of this campaign: it is a previously released trace corpus of the same benchmark joined by problem identifier. Simulated from discordance rates in that corpus, the paired per-actor main effect is well powered; a post-hoc detectability analysis finds the actor × intervention interaction powered only for gpt-oss against gemma-4 (§4.1). An early campaign stopped for a truncation confound was never pooled (supplement).

The feedback-matched condition. It was run after the freeze (4,748 trajectories) on the same endpoint, problems, budget and evaluator, for all three actors on Omni-MATH and for gemma-4 and gpt-oss on SciBench [38], manifested separately and never pooled into a frozen cell; the two gemma-4 cells were run a second time, independently. A re-run of gemma-4's role-structured condition on 400 frozen problems finds no detectable drift in final accuracy (0.818 against 0.825 frozen, paired $p = 0.75$; recovery 0.198 against 0.146). Decoding at temperature 0 is not deterministic on a shared endpoint: the two role-free conditions, byte-identical until the first verdict, agree on gemma-4's first verdict for 1,496 of 1,598 problems, and their first-pass collapse differs by up to 0.04 (Table 1). We therefore treat a cell as one sample and give the second run, where we have one, in the supplement. Prompts, runner and analysis scripts are in the supplementary material.

5 Results

All contrasts are paired at the problem level with bootstrap intervals clustered by problem; trajectories never producing a graded answer count as incorrect (the first-pass contrast of §5.1 excepted), and significance is exact McNemar on discordant pairs, with p to two significant figures.

5.1 R1. Removing roles changes the organization, not the first answer

Besides deleting the role clauses, the role-free prompt asks every agent for its current best answer; agents, order, budget and evaluator are unchanged. We measure whether *every* agent asserts its own final answer on the first pass, where trajectory length cannot confound it (Table 1): the rate rises from 0.000–0.120 to 0.846–0.976. **Decomposed by role, the planner supplies most of the**

Table 1: Functional collapse, Omni-MATH. First pipeline passes in which all three agents assert their own final answer, with exact byte duplication (a *lower bound*: precision 1.000, recall 0.262) and a model adjudication that is outcome-blind but not condition-blind. Collapse is not applicable to the enforced conditions, whose schema has no answer field; for the role-free condition with feedback only collapse was measured.

actor	functional collapse	exact byte duplication	adjudicated no division
role-structured			
gemma-4-31B	0.000	0.000	0.000
gpt-oss-120b	0.120	0.000	0.000
Llama-3.3-70B	0.000	0.000	0.000
role-free, feedback withheld			
gemma-4-31B	**0.976**	0.000	**0.857**
gpt-oss-120b	**0.846**	0.680†	**1.000**
Llama-3.3-70B	**0.872**	0.008	**1.000**
role-free, with feedback			
gemma-4-31B	**0.974**	–	–
gpt-oss-120b	**0.883**	–	–
Llama-3.3-70B	**0.900**	–	–
structurally enforced			
all actors	n/a	≤ 0.001	0.071–0.188

Sensitivity. Under three answer-marker definitions the feedback-withheld role-free rate is 0.846–0.999; the role-structured rate is 0.000–0.238 under the two stricter markers and 0.328–0.892 under the loosest, which also counts any "therefore", "hence" or "= number" (as when a Planner restates givens), so read the role-structured level as "low," not a point estimate. The ordering holds in all 9 of 9 actor×marker cells.
†Not interpretable: 80% of this cell's planner messages are under 500 visible characters (the logging defect, §5.1).

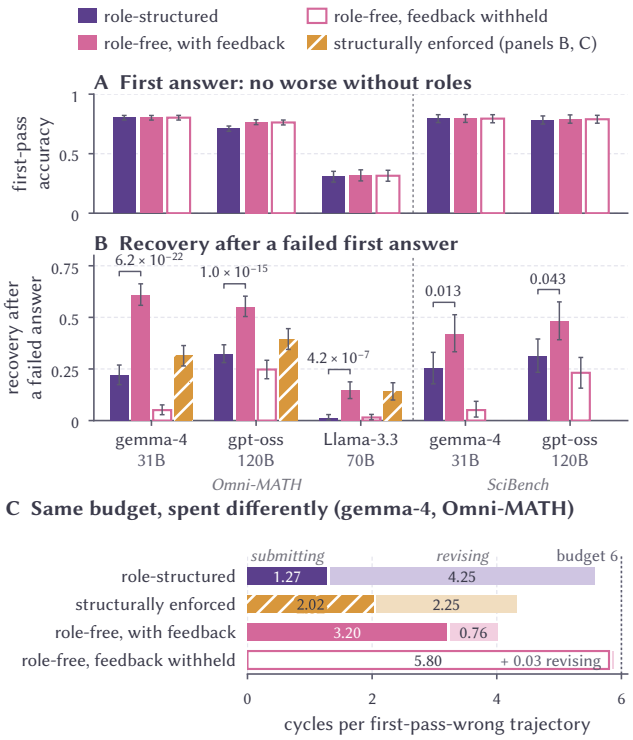


Figure 1: Roles organize the team; given the same feedback, they do not add recovery. (A) Removing roles leaves first-pass accuracy unchanged or higher. (B) Recovery after a failed first answer. Given the same feedback string, the role-free team recovers more than the role-structured one; the role-free team with the feedback withheld (hollow) recovers least, tying the role-structured team for Llama-3.3. Bars are *marginal* rates over each condition's own first-pass-wrong set; brackets give the *paired* exact McNemar p, role-structured versus role-free with feedback. Enforcement was run on Omni-MATH only. (C) How gemma-4's first-pass-wrong Omni-MATH trajectories spend the six-cycle budget: cycles ending in a submission (approval) or in a revision (a decline). (A) is over trajectories with a first verdict; in (B) those that never finish count as unrecovered; bootstrap 95% intervals.

change: the executor answers in both conditions and the reviewer, already high for two actors, rises in all three, but the planner moves 0.000/0.183/0.000 to 0.979/0.898/0.889 (gemma-4 / gpt-oss / Llama-3.3). Since the footer asks for an answer, this measures what a Planner does once its MUST NOT is gone, not an emergent tendency: the claim is about the *change*, not the level.

Content duplication and a second opinion. gemma-4's and Llama-3.3's role-free planners write full independent solutions (≈1,880 characters, median) before boxing an answer, and numeric-token overlap with the executor exceeds its per-condition null by +0.449 and +0.432 (+0.159 and +0.145 with roles). A frontier model judging 171 trajectories against a frozen rubric agrees (Table 1, last column; repeat agreement 0.953), but it is outcome-blind, not condition-blind: role words were renamed rather than equalized, and a renamed token appears in all 42 role-structured items and 3 of 56 role-free ones (Fisher $p = 1.5 \times 10^{-24}$). The collapse claim does not rest on it.

First-pass accuracy does not follow. On problems graded under both conditions (n = 1,600 / 1,592 / 389) first-pass accuracy is *identical* for gemma-4 (+0.0000, [−0.0156, +0.0150]) and flat for Llama-3.3 (+0.0077), while it *improves* for gpt-oss (+0.0503, [+0.0339, +0.0666]; Figure 1A). Whatever roles buy, they do not buy the first answer.

A logging defect. Our runner logged each response's `content`, falling back to `reasoning` only when content was empty. gpt-oss-120b often returns a short content string beside a long hidden trace, so its derivations are missing from the record (75.7% of its role-free messages, 93.4% of its first passes; the flag meant to report this reports 5.8%). Agents pass content only, by design and in every condition, so the defect is in the record: no point estimate of gpt-oss's duplication is recoverable. It does not manufacture the collapse: on first passes with no affected message, role-free collapse is 0.821 (n = 106) and role-structured 0.118.

5.2 R2. Given the same feedback, the role-free team recovers more

What a cycle is. The pipeline runs a fixed grid of six deliberation cycles — three outer rounds of two inner rounds — each one pass of Planner → Executor → Reviewer. The Reviewer decides whether a cycle ends in a *submission* or a *revision*. On approval the current answer goes to the evaluator, which returns PASS — the trajectory ends — or FAIL with a repair hint; without approval nothing is submitted and the next cycle carries the Reviewer's critique forward. A trajectory that never submits is graded once when the budget runs out.

Recovery and final accuracy. The role-structured team receives the evaluator's feedback, so the like-for-like comparison is with the role-free team that receives it too (Table 2). **That team recovers more, not less**: 0.609 against 0.222 (marginal, Figure 1B), and on the 236 problems both first answer wrongly, 89 are recovered only without roles and 4 only with them ($p = 6.2 \times 10^{-22}$). Both reach the same first-pass rate and a correct first answer is never lost, so this becomes final accuracy: 0.921 against 0.846, with 137 problems correct only without roles and 17 only with them ($p = 1.7 \times 10^{-24}$, conditioned on nothing). The recoveries are revisions, not re-gradings: in 192 of the 193 the passing answer differs from the one first rejected (541 of 547 across the five cells with feedback matched), and repeated identical evaluator prompts disagree in only 0.0072 of cases.

Reproducibility, drift and leakage. The two gemma-4 feedback-matched cells were run twice, independently and with identical code. The second run is close to the first (Omni-MATH recovery 0.586 against 0.609, final accuracy 0.922 against 0.921; SciBench recovery 0.435 against 0.419), but SciBench's paired test weakens from $p = 0.013$ to 0.052; we report the first throughout (supplement). A re-run of the role-structured condition shows no detectable drift in final accuracy (§4.2). Repair hints contain the gold answer verbatim in at most 2.3% of failing verdicts in any cell; for gemma-4 on Omni-MATH, dropping every trajectory that saw one leaves 0.609 and 0.222 at 0.599 and 0.212. We did not audit partial leakage.

Scope across actors. Both other actors repeat the ordering in recovery and in final accuracy (Table 2), and in both the feedback leaves first-pass accuracy where it was (0.765 and 0.317, against 0.762 and 0.315 with it withheld). Llama-3.3's role-structured team rarely submits at all, and there enforcement recovers about as much as the role-free team with feedback (0.139 against 0.147 marginally; paired 9 against 17 of 219, $p = 0.17$). Against the feedback-withheld baseline the role-structured advantage is clean only for gemma-4: it is directional for gpt-oss (0.324 against 0.247 marginally; paired 52 against 39 of 327, $p = 0.21$) and absent for Llama-3.3 (0.014 against 0.015). gpt-oss recovers without feedback, consistent with its answer moving from cycle to cycle anyway (one distinct answer in 20.8% of its failed trajectories, against 84.5% for gemma-4).

5.3 R3. Matched on verdicts, the gap closes; roles gate submission

The feedback-withheld baseline. Without the feedback, the role-free team turns nearly its whole budget into submissions: across gemma-4's first-pass-wrong trajectories it submits in 5.80 of 5.83 cycles (Figure 1C), its third agent never declines in 98.1% of them, and in 84.5% the entire transcript holds exactly one distinct answer. Nothing in its prompts carries the feedback: each FAIL is followed by another cycle over the same history. Given the verdict and hint, the identical prompts produce a different team: the answer moves (one distinct answer in 6.6% of failed trajectories) and recovery rises from $16/316 = 0.051$ to $193/317 = 0.609$; of the 265 problems both miss first, 139 are recovered only with the feedback and 2 only without ($p = 7.2 \times 10^{-39}$). First-pass accuracy is 0.80 in both conditions. Of the 16 recoveries without feedback, 7 are the same answer graded again. **For gemma-4, revision needs the feedback, not the roles.**

Matched on verdicts, the gap closes. The role-free team with feedback also receives more of it: each submission returns a verdict and, on a FAIL, a hint written with the gold answer in view. To compare the two teams at equal exposure (a post-hoc analysis), we take the problems both first answer wrongly and cut each role-free trajectory off after as many verdicts as the role-structured team received on the same problem. No cell shows a significant difference (Table 2): 0.165 against 0.153 for gemma-4 ($p = 0.69$), 0.229 against 0.216 for gpt-oss ($p = 0.72$), and 0.230 against 0.190 and 0.215 against 0.247 on SciBench ($p \geq 0.48$); Llama-3.3's role-structured team submits so rarely that its cell has five events. Pooled and stratified by cell, the difference is +0.009 in favor of roles (95% paired bootstrap CI −0.014 to +0.033). The match is coarse: where the role-structured team received one verdict the pair ties at zero by construction (141 of gemma-4's 236 pairs). Restricted to problems where it received at least two, the difference is +0.018 (−0.029 to +0.065); cutting both trajectories at the smaller verdict count gives −0.005; recoding the scoring defect of §7 gives +0.029 (+0.004 to +0.054; supplement). The comparison crosses the freeze, so we repeat it against the role-structured re-run made after it: on the 65 problems both first answer wrongly, recovery is 3 against 22 in favor of the role-free team unmatched ($p = 1.6 \times 10^{-4}$) and 4 against 1 verdict-matched ($p = 0.38$). No variant puts the role-free team significantly ahead verdict for verdict; the only significant variant favors roles, by a small margin. The role-free team recovers more because it asks for more verdicts.

What roles do instead: a gate on submission. The gap in verdicts comes from the role-structured Reviewer. Across gemma-4's failed trajectories that team submits 1.27 times in 5.52 cycles, against 3.20 in 3.96 without roles (Figure 1C). Of its declining cycles, 53% come before the first submission, when no feedback exists, and 84 of its 316 failed trajectories never submit within the budget: they are graded once at the end and never see the feedback. One clause forbids approval in a cycle that receives a FAIL, but it covers only 19% of the declines; in the later cycles that receive no FAIL, where approval is allowed, the Reviewer still declines in 75% of 504 such cycles, against 4% of 184 for the role-free team's third agent. On Omni-MATH the clause covers 37% of gpt-oss's declines and 4% of Llama-3.3's, whose role-structured team submits 0.32 times per failed trajectory and never submits in 205 of 276; on SciBench it covers 40% and 60% (supplement). The gate looks like precision — recoveries per verdict after the first are higher with roles (0.41 against 0.28 for gemma-4) — but that compares different

Table 2: The feedback-matched contrast in every cell. RS: role-structured; RF: role-free with feedback; RF$^-$: role-free, feedback withheld. Recovery is marginal, over each condition's own first-pass-wrong trajectories. The paired columns use the problems both RS and RF first answer wrongly (n): those recovered by only one of them, with the exact McNemar p; and recovery when each RF trajectory is cut off after as many evaluator verdicts as RS received on the same problem, so that both receive the same number of verdicts and hints. Final accuracy is over all problems, paired, conditioned on nothing (never-graded trajectories count as incorrect). Submissions are approvals per first-pass-wrong trajectory; the last column counts RS's first-pass-wrong trajectories that never submitted in six cycles, graded once when the budget ran out.

	recovery			recovered only by (paired)				verdict-matched			final accuracy			submissions		RS never
actor, domain (n)	RS	RF	RF$^-$	n	RS	RF	p	RS	RF	p	RS	RF	p	RS	RF	submits
gemma-4, Omni-MATH (1,600)	0.222	0.609	0.051	236	4	89	6.2×10^{-22}	0.165	0.153	0.69	0.846	0.921	1.7×10^{-24}	1.27	3.20	84/316
gpt-oss, Omni-MATH (1,600)	0.324	0.552	0.247	319	17	101	1.0×10^{-15}	0.229	0.216	0.72	0.803	0.894	2.7×10^{-28}	2.10	3.53	51/463
Llama-3.3, Omni-MATH (400)	0.014	0.147	0.015	250	1	26	4.2×10^{-7}	0.012	0.008	1.0	0.315	0.415	1.6×10^{-7}	0.32	2.71	205/276
gemma-4, SciBench (574)	0.254	0.419	0.051	100	7	21	0.013	0.230	0.190	0.48	0.847	0.880	0.0054	1.95	3.62	3/118
gpt-oss, SciBench (574)	0.315	0.483	0.231	93	7	18	0.043	0.215	0.247	0.68	0.852	0.892	1.9×10^{-4}	2.68	3.95	1/124

trajectories; on the same problem and the same number of verdicts, the role-free team is no better.

Removing the gate alone. The direct test keeps the role-structured prompts byte for byte but submits the team's answer in every cycle, whatever the Reviewer decides (exploratory, one run: gemma-4 on the 400 Omni-MATH problems of the post-freeze re-run; supplement). First-pass accuracy is unchanged (0.768 against 0.773), the team draws 4.20 verdicts per failed trajectory against 1.44, and recovery rises from 0.198 to 0.613, as high as the role-free team's 0.549 on the same problems. On problems both first answer wrongly, the forced team recovers 26 that the gated one misses and 2 the other way ($p = 3.0 \times 10^{-6}$); against the role-free team the split is 7 to 7 ($p = 1$). With the role contract intact, removing the gate removed the gap: against a ground-truth evaluator and six cycles, a withheld submission is a lost chance to recover.

5.4 R4. The same ordering in a second domain

SciBench's complete text-only slice of 574 free-response college science problems, frozen separately, tests whether the pattern belongs to the organization or to competition mathematics. The role-free prompt is byte-identical to its Omni-MATH form; the role-structured one keeps the same structural contract with domain wording adapted, as does the evaluator. For both actors the ordering returns (Table 2): first pass is flat, recovery is highest with feedback and no roles (0.419 and 0.483 against 0.254 and 0.315 with roles), and final accuracy is higher without roles ($p = 0.0054$ and 1.9×10^{-4}). The recovery contrast is weaker ($p = 0.013$ and 0.043), and as on Omni-MATH it vanishes when both teams are credited with the same number of verdicts. One part does not transfer: gemma-4's role-structured answer does not lock (one distinct answer in 7.6% of failed trajectories, against 8.5% for the role-free team with feedback), and by the second verdict the two recover alike (0.237 against 0.239; supplement).

5.5 R5. Structural enforcement shifts the allocation, and introduces brittleness

Enforced and role-structured prompts carry the same advice slot, so this contrast is matched on feedback. Enforcement replaces the request with a constraint: the Planner's answer becomes unrepresentable. The manipulation check passes: the planner asserts a final answer in 0.000 of enforced first passes in all three actors against 0.979/0.898/0.889 without roles, with 838/1,183/395 role-boundary rejections per cell. It does not change what the planner *computes*, which is why the adjudication still finds no division in 0.071–0.188 of enforced first passes: an agent can be structurally stopped from answering without being stopped from doing the work. Relative to the role-structured team, enforcement loosens the gate: it submits more often (2.02/2.33/1.84 against 1.27/2.10/0.32 per failed trajectory) and, for gemma-4 and Llama-3.3, **unlocks the answer** (one distinct answer in 0.119 and 0.072 of failed trajectories against 0.288 and 0.750; for gpt-oss 0.092 against 0.052).

Recovery follows most clearly for the weakest actor: Llama-3.3's rises about tenfold, from 0.014 to 0.139 ($p = 3.8\times10^{-6}$); for gemma-4 (0.314 against 0.222, $p = 0.30$) and gpt-oss (0.395 against 0.324, $p = 0.11$) the direction is the same but not resolvable against attrition. Recovery is computed on every trajectory with a recorded incorrect first verdict, including the 150 enforced trajectories that received one then died in regeneration, scored 0 by the pre-registration; conditioning instead on successful grading would manufacture significance out of one-sided attrition.

Improved revisability does not become end-to-end accuracy, because enforcement is brittle. Under the pre-registered rule the accuracy difference is −0.0225 / −0.0050 / +0.0175 (gemma-4 / gpt-oss / Llama-3.3): one significant negative, two nulls. Dropping the 8.0%/9.9%/17.0% of enforced trajectories that never finish — against 0.0–0.8% role-structured and 0.0–2.5% role-free — gives +0.0190 / +0.0361 / +0.0453: all positive, all significant, ordered by actor weakness, exactly the published capability-equalizer direction [2]. **That second reading is survivorship**: the pre-registration fixed in advance that such trajectories count incorrect. Restoring shared history at fixed enforcement raises gemma-4's pre-registered accuracy (+0.0450, $p = 0.0079$; as graded +0.0196, $p = 0.14$) but not gpt-oss's (−0.0125, $p = 0.56$). Both scorings with intervals are in the supplement.

6 Discussion

What roles do after feedback. Producing a first answer is a *generation* problem: three agents reaching the same place by three routes

cost nothing but tokens, and first-pass accuracy is nearly indifferent to the division of labor. After a failure the team must produce something unlike the incumbent candidate, and, gpt-oss aside, it rarely does so unless it knows the candidate failed. Given that knowledge, the role contract adds little capacity to revise: verdict for verdict, the role-free team is no better. What the contract decides is *when* the team asks: the role-structured Reviewer withholds approval until it is satisfied, so the team draws fewer verdicts, sometimes none. Overriding the Reviewer closed the gap for gemma-4. Under a ground-truth evaluator and a fixed cycle budget, fewer verdicts mean fewer recoveries. Where verification is costly, noisy or unavailable, the usual case outside a benchmark, a team that asks less could be the better design; this experiment cannot say. In the terms of §2, the gate lives in the deontic part of the role contract — when the Reviewer may approve — and deleting the roles deleted it together with the functional assignment.

What a builder can take from this. Two things. With a cheap, reliable verifier, a Reviewer role that withholds submission costs recovery: let the team submit, or gate on the verifier rather than on the critic. And whatever the roles, route the verifier's feedback to every agent that can act on it: in our pipeline the gemma-4 and Llama-3.3 teams without that slot almost never recovered, and the same prompts with it recovered most.

What a role ablation removes. In a prompted multi-agent system the channels through which agents receive feedback, tool output or a verifier's verdict are slots in the same templates that carry the roles. Rewriting the templates to remove the roles can silently remove a channel with them: nothing in the budget record, the call counts or the accuracy reports its absence. A role ablation therefore needs an information audit beside the budget audit — for each agent and condition, the set of runtime inputs its template actually consumes — and a contrast is matched only where those sets agree. Ours differ in one input: the feedback.

What a role diagram is evidence of. Precise reviewing need not mean critique is taken up [40], and a clean division of labor need not mean a team adapts; both are intermediate properties, and what matters is whether corrective information changes the next action. A diagram is a hypothesis, and task-outcome evaluation — the only evaluation most multi-agent systems receive — cannot test it: an aggregate score averages a stage where the organization is idle with one where it does all its work. Our gemma-4 team lost its division of labor and moved final accuracy by −0.034 with the feedback withheld and +0.075 with it, both almost entirely after a first answer had failed. Two cheap additions expose this: report performance *conditioned on the first attempt having failed*, and *measure* the realized organization rather than assume the specified one.

7 Limitations

The recovery loop is oracle-gated. The evaluator triggering retries sees ground truth, so every recovery number is recovery *given* a ground-truth-informed failure signal: not unaided self-correction, and stop-on-correct censors destructive revision. An evaluator that can be queried up to six times favors the team that queries it most; matched on verdicts, the role-free advantage disappears (§5.3).

The feedback-matched condition. It was added after the snapshot was frozen and is outside the pre-registration; a re-run of gemma-4's role-structured condition on 400 problems shows no detectable drift in final accuracy, but the conditions come from separate runs. It matches the role-structured condition on the feedback string, not on its framing: role-structured agents receive it inside role-specific instructions, role-free agents as a bare `[Evaluator]` entry. The forced-submission run (§5.3) is post-freeze too, covers one actor, one domain and one run, and removes the whole gate rather than the one clause that forbids approval after a FAIL.

A scoring defect, which narrows the gap. Beyond the logging defect of §5.1, the runner grades the last answer *submitted* but records the last answer *held*; for gemma-4 these disagree on 17% of role-structured, 0% of feedback-withheld and 13% of feedback-matched role-free first-pass-wrong trajectories (gpt-oss 16%/7%, Llama-3.3 6%/0.4%, role-structured/feedback-withheld). Recoding the trajectories whose held answer matched the key but was graded wrong (62 in the nine main frozen Omni-MATH cells, against 0 of the snapshot's 10,734 graded trajectories whose two answers agree; 3 in gemma-4's feedback-matched run) moves gemma-4's role-structured recovery to 0.275 and the feedback-matched role-free one to 0.618: the gap narrows from 0.387 to 0.343. We keep the scoring rule fixed before the data as primary.

Scope, provenance, and blind spots. Enforcement's 8–17% one-sided attrition (§5.5) is why its as-graded gain is survivorship and why the context-restored condition is only a *partial* control. We test one PER organization, with an evaluator that is itself one of the actors under study, and the second domain adapts the role-structured and evaluator prompts (§5.4). The functional measure detects that every agent asserted an answer, not that their reasoning was the same.

8 Conclusion

Multi-agent roles organize a team: deleting them destroyed the intended division of labor in all three actors on Omni-MATH, while first-pass accuracy did not fall. They looked load-bearing for repairing a rejected answer only against a role-free team that never received the evaluator's feedback. Given the same feedback, the role-free team recovered more in every cell (paired $p = 6.2 \times 10^{-22}$ for gemma-4 on Omni-MATH); matched on the number of verdicts, that advantage disappeared (pooled difference +0.009 in favor of roles, 95% CI −0.014 to +0.033). Roles changed little in how a team uses corrective feedback and much in how often it asks for it. In one exploratory gemma-4 run, submitting in every cycle gave the role-structured team a recovery as high as the role-free team's. **Structure is not adaptability: evaluating agent organization means asking not only whether the specified division of labor is realized, but whether corrective information reaches the team and changes what it does next.**

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